

Login Attempt Analyzer

Mini Project Report for Avishkar Andhra Internship

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Login Attempt Analyzer

1. Abstract

The Login Attempt Analyzer is a cybersecurity-based mini project developed using Python, Pandas, and Matplotlib. The project analyzes login activity records and identifies suspicious login attempts. It provides useful insights regarding successful logins, failed logins, attack attempts, and account takeover incidents.

The project helps in understanding user behavior and improving system security through data analysis and visualization. A dashboard is generated to provide graphical representation of login activities.

2. Introduction

Cybersecurity has become an important aspect of modern information systems. Monitoring login activities helps organizations detect unauthorized access and suspicious behavior.

This project was chosen to understand log analysis techniques and to demonstrate how Python can be used for security monitoring. The Login Attempt Analyzer processes login records and generates meaningful insights using graphical dashboards.

3. Objective

- Analyze login records from a dataset.
- Count successful and failed login attempts.
- Identify suspicious attack activities.
- Detect account takeover incidents.
- Generate graphical dashboards for visualization.

4. Technologies Used

- Python
- Pandas
- Matplotlib
- Jupyter Notebook (Anaconda)

5. Dataset Used

Dataset Link: [rba-dataset.csv](#)

The project uses a login activity dataset containing details such as:

Login Attempt Analyzer

- Login Timestamp
- User ID
- IP Address
- Country
- Device Type
- Browser Information
- Login Status
- Attack Indicators

6. Methodology

Step 1: Load the dataset using Pandas.

Step 2: Perform data exploration and understand the columns.

Step 3: Calculate total login events.

Step 4: Count successful and failed login attempts.

Step 5: Identify attack IP addresses and account takeover events.

Step 6: Perform country-wise and device-wise analysis.

Step 7: Generate visualizations and create a dashboard using Matplotlib.

Step 8: Save charts and interpret the results.

7. Source Code

Example:

Project Folder

```
|  
|— Fraud_Detection.ipynb  
|— rba-dataset.csv  
|— Balu_Login_Attempt.txt  
|— Screenshots/reports  
|— README.txt
```

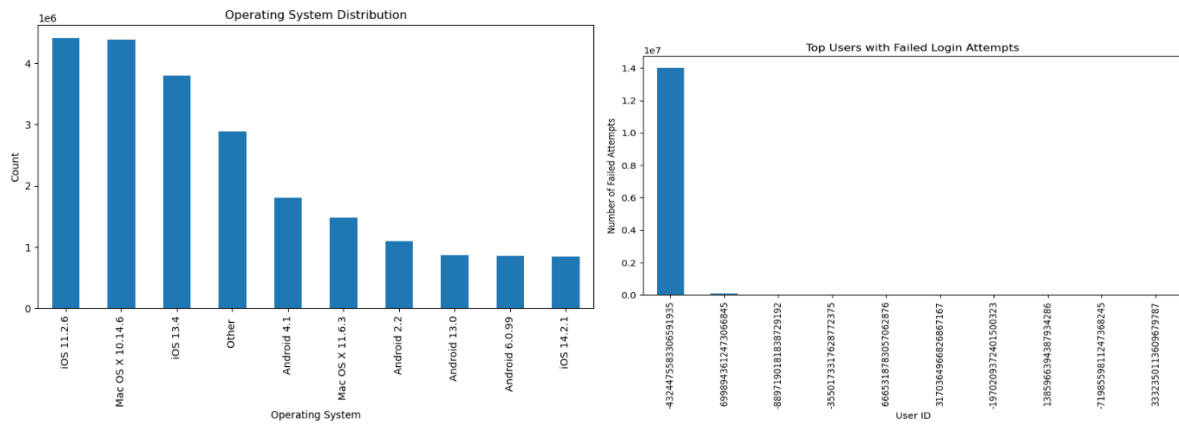
GitHub Link: [Login-Attempt-Analyser](#)

Login Attempt Analyzer

8. Output Screenshots

	index	Login Timestamp	User ID	Round-Trip Time [ms]	IP Address	Country	Region	City	ASN	User Agent String	Browser Name and Version	OS Name and Version	Device Type	Login Successful
0	0	2020-02-03 12:43:30.772	-4324475583306591935	NaN	10.0.65.171	NO	-	-	29695	Mozilla/5.0 (iPhone; CPU iPhone OS 13_4 like ...	Firefox 20.0.0.1618	iOS 13.4	mobile	False
1	1	2020-02-03 12:43:43.549	-4324475583306591935	NaN	194.87.207.6	AU	-	-	60117	Mozilla/5.0 (Linux; Android 4.1; Galaxy Nexus...	Chrome Mobile 46.0.2490	Android 4.1	mobile	False
2	2	2020-02-03 12:43:55.873	-3284137479262433373	NaN	81.167.144.58	NO	Vestland	Urangsvag	29695	Mozilla/5.0 (iPad; CPU OS 7_1 like Mac OS X) ...	Android 2.3.3.2672	iOS 7.1	mobile	True
3	3	2020-02-03 12:43:56.180	-4324475583306591935	NaN	170.39.78.152	US	-	-	393398	Mozilla/5.0 (Linux; Android 4.1; Galaxy Nexus...	Chrome Mobile WebView 85.0.4183	Android 4.1	mobile	False
4	4	2020-02-03 12:43:59.396	-4618854071942621186	NaN	10.0.0.47	US	Virginia	Ashburn	398986	Mozilla/5.0 (Linux; U; Android 2.2) Build/NMA...	Chrome Mobile WebView 85.0.4183	Android 2.2	mobile	False

Fig: Dataset



Login Attempt Analyzer

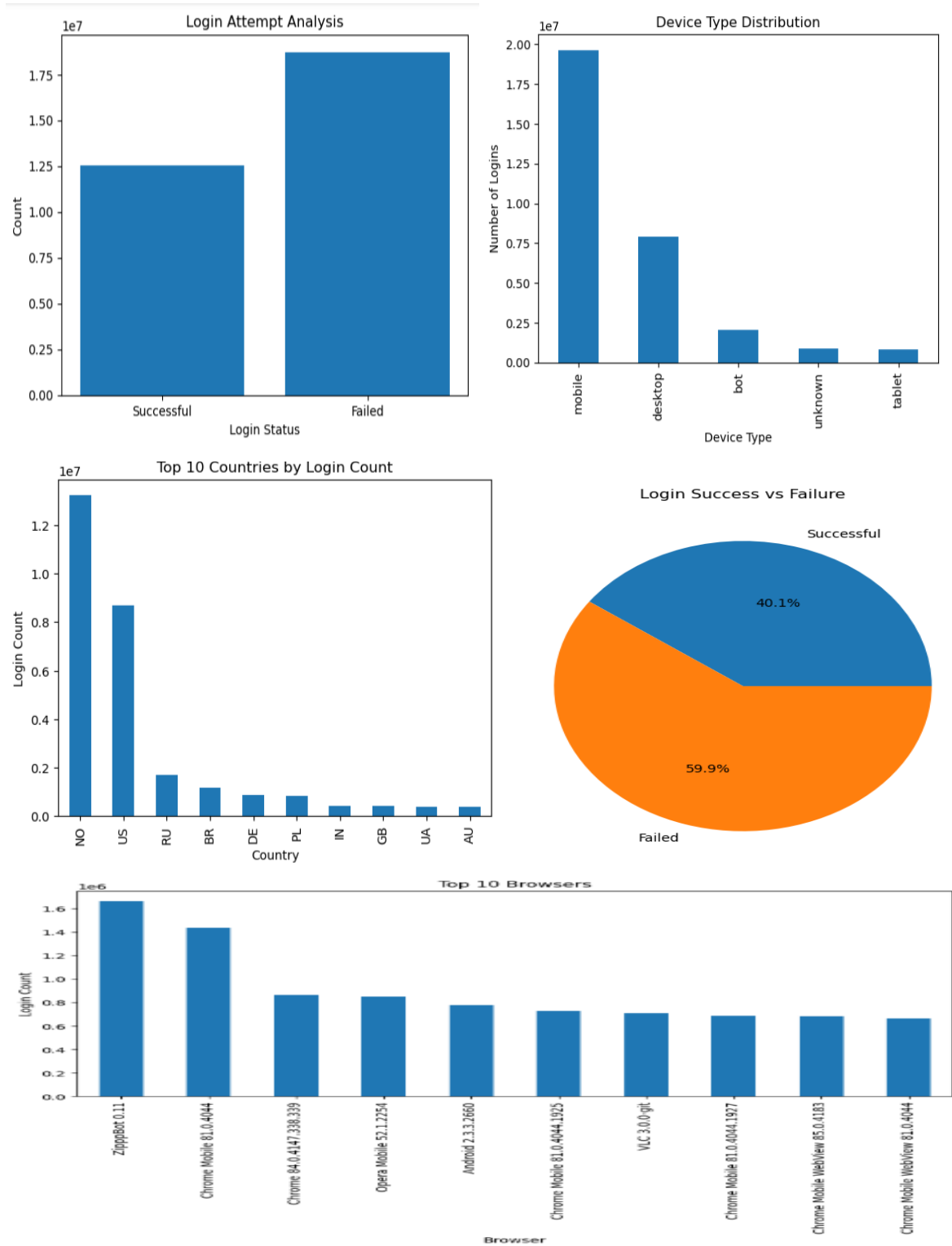


Fig: Individual Charts

Login Attempt Analyzer



Fig: Dashboard with all charts

9. Conclusion

The Login Attempt Analyzer successfully analysed login records and identified suspicious activities. The project provided insights into successful logins, failed logins, attack attempts, and account takeover events.

Through this project, knowledge of Python, Pandas, data analysis, and visualization techniques was improved. The generated dashboard helps in understanding login behaviour and enhancing cybersecurity monitoring.